PRACTICE PROBLEMS WEEK-3

Ques.1 Find an example of a d-system which not a σ-field.

Solve: Let $\mathcal{C}$ = { countable disjoint union of $(a,b]$: $a,b \in \mathbb{R}$ }

Now:

$$R = (-\infty, \infty] = (-\infty, \infty) \in \mathcal{C}$$

$\Rightarrow$ i) $\mathcal{U} = \mathcal{C}$

ii) if $(a_1, b_1] \cup (a_2, b_2] \in \mathcal{C}$ be union:

Say $b_1 < a_2$

Then $(a_1, b_1]^c = (-\infty, a_1] \cup (b_1, \infty)$

$(a_2, b_2]^c = (-\infty, a_2] \cup (b_2, \infty)$

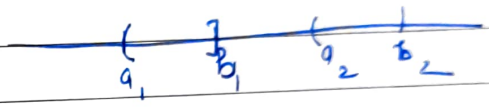
and

$$\left((a_1, b_1]^c \cap (a_2, b_2]^c\right) = (-\infty, a_1] \cup (b_1, a_2] \cup (b_2, \infty)$$

Say if $A = A_1 \cup A_2$

$A_1 = (a_1, b_1]$

$A_2 = (a_2, b_2]^c$



Then

$$A^c = A_1^c \cap A_2 \in \mathcal{C}$$

Similarly if $A = \bigcup_{i=1}^{n} A_i$ for finite n:

$$A^c = A_1^c \cap A_2^c \cap \dots \cap A_n^c \in \mathcal{C}.$$

iii)

Now, let $A_n = \left(0, 1 - \frac{1}{n}\right] \in \mathcal{C} \ \forall n \in \mathbb{N}$

but $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} \left(0, 1 - \frac{1}{n}\right] = (0,1) \notin \mathcal{C}$

So $\mathcal{C}$ is NOT a σ-field.

but for $A \ B_n = \bigcup_{i=1}^{k} (a_n^i, b_n^i]$ where B_n is disjoint union of disjoint type $(a_n^i, b_n^i]$

now $\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} \left(\bigcup_{i=1}^{k} (a_n^i, b_n^i] \right) \in \mathcal{C}$ elements

(∵ This is countable union of $(a_n^i, b_n^i]$'s).

Q.2. Find an example of a Monotone-class which is not a field.

Solve: Let $\Omega = \{1, 2, 3\}$

$\mathcal{C} = \{\{1\}, \{2\}, \{1,2\}\}$

Now,

for $A_1, A_2 \ldots A_n \in \mathcal{C}$

we see if $A_1 \supseteq A_2 \supseteq A_3 \ldots$

Then $\bigcap_{i=1}^{\infty} A_i \in \mathcal{C}$

& if $A_1 \subseteq A_2 \subseteq A_3 \subseteq \ldots$

$\bigcup_{i=1}^{\infty} A_i \in \mathcal{C}$

$\Rightarrow$ $\mathcal{C}$ is a monotone class

but $\Omega \notin \mathcal{C}$

So $\mathcal{C}$ is NOT a field.

Ques 3. a) If a collection $\mathcal{F}$ of subsets is both a π-system and a d-system, then show that it a σ-field.

Solve: by definition, if $\mathcal{C} \in 2^{\Omega}$ s.t. $\mathcal{C}$ is π-systems and d-systems. then i), ii) by d-system.

Now:

Let $A_1, A_2, \ldots A_n \in \mathcal{C}$.

define

$B_1 = A_1$

$B_2 = A_2 / (A_1 \cup A_2)$ ($\because A_1, A_2 \in \mathcal{C}$ $\Rightarrow A_1^c, A_2^c \in \mathcal{C}$ (d-Sysstm) $\Rightarrow A_1^c \cap A_2^c \in \mathcal{C}$ (π-systm) $\Rightarrow (A_1 \cup A_2)^c \in \mathcal{C}$ (d-system) $\Rightarrow (A_1 \cup A_2) \in \mathcal{C}$ (d-system)

Similarly we see

for $A_1, A_2, \ldots A_n \in \mathcal{C}$

$A_1 \cup A_2 \ldots \cup A_n \in \mathcal{C}$.

So

Let us define

$B_n = A_n / (A_1 \cup A_2 \cup \ldots A_{n-1})$

As per we see that (In Ist assignment) that

$$\bigcup_{i=1}^{\infty} B_i = \bigcup_{i=1}^{\infty} A_i$$

Since $\bigcup_{i=1}^{\infty} B_i \in \mathcal{C}_e$ ($\because \mathcal{C}_e$ is d-system and B_i are all disjoint

So $\bigcup_{i=1}^{\infty} A_i \in \mathcal{C}_e \Rightarrow \mathcal{C}_e$ is a σ-field. =

b) Given F is a Monotone Class, and Field. but

$A_1, A_2, A_3, \ldots A_n \ldots \in F$ be a countable Collection of sets of F.

Then define

$$B_1 = A_1; \quad B_2 = A_1 \cup A_2, \ldots$$
$$B_n = \bigcup_{i=1}^{n} A_n \qquad \forall\, n \in \mathbb{N}.$$

Since each B_i is finite union of A_i

$\Rightarrow B_i \in F \quad \forall\, i \in \mathbb{N}$.

Now $B_1 \subseteq B_2 \subseteq B_3 \subseteq \ldots B_n \subseteq \ldots$

Since, F is, monotone class

So $\bigcup_{i=1}^{\infty} B_i \in F$

and we note that $\bigcup_{i=1}^{\infty} B_i = \bigcup_{i=1}^{\infty} A_i$

$\Rightarrow \bigcup_{i=1}^{\infty} A_i \in F$

$\Rightarrow$ F is a σ-field.

Quiz 4.

1) F - π-system on Ω

$P_1, P_2 : (\Omega, \sigma(F)) \longrightarrow [0,1]$

s.t.

$$P_1(A) = P_2(A), \quad \forall A \in F$$

Now:

$$-P_1(A) = -P_2(A) \quad \forall A \in F$$

$$\Rightarrow 1 - P_1(A) = 1 - P_2(A) \quad \forall A \in F$$

$$\Rightarrow P_1(A^c) \in P_2(A^c) \quad \forall A \in F.$$

Let $A_1, A_2 \in F \Rightarrow A_1 \cap A_2 \in F$ ($\because F$ is π-system)

So ~~$P(A_1 \cap A_2) = P_2(A_1 \cap A_2)$~~

$\Rightarrow$ ~~$P(A_1 \cap A_2$~~

$$P_1(A_1^c) = P_1(A_1^c) \quad \because A_1, A_2 \in F$$

and $$P_2(A_2^c) = P_2(A_2^c)$$

Now: $\Rightarrow$

$$P_1(A_1 \cap A_2) = P_2(A_1 \cap A_2)$$

$$\Rightarrow P_1((A_1 \cap A_2)^c) = P_2((A_1 \cap A_2)^c)$$

$$\Rightarrow P_1(A_1^c \cup A_2^c) = P_2(A_1^c \cup A_2^c)$$

$$\Rightarrow P_1(A_1^c) + P_1(A_2^c) - P_1(A_1^c \cap A_2^c) = P_2(A_1^c) + P_2(A_2^c) - \cancel{P_2(A_1^c)} - P_2(A_1^c \cap A_2^c)$$

$\Rightarrow$ ~~$P(A_1^c)$~~

$$P_1(A_1^c \cap A_2^c) = P_2(A_1^c \cap A_2^c)$$

$$\Rightarrow P_1((A_1 \cup A_2)^c) = P_2((A_1 \cup A_2)^c)$$

$$\Rightarrow P_1(A_1 \cup A_2) = P_2(A_1 \cup A_2),$$

Similarly for A_1, A_2, A_3. By PMI by induction we can prove for countable set. Since each $A \in \sigma(F)$ can be written as countable union, intersection or complements of F. So

$$P_1(A) = P_2(A) \quad \forall A \in \sigma(F).$$

b) Now; let $\mathcal{F}$ be a field on Ω.

$\Rightarrow A_1, A_2, \ldots A_n \in \mathcal{F}$

$\Rightarrow A_1^c, A_2^c, \ldots A_n^c \in \mathcal{F}$

$\Rightarrow A_1^c \cup A_2^c \cup \ldots \cup A_n^c \in \mathcal{F}$

$\Rightarrow (A_1^c \cup A_2^c \cup \ldots \cup A_n^c)^c \in \mathcal{F}$

$\Rightarrow A_1 \cap A_2 \ldots \cap A_n \in \mathcal{F}$.

$\Rightarrow \mathcal{F}$ is a π-system

As proved in part Part a):

If $P_1(A) = P_2(A) \quad \forall A \in \mathcal{F}$

$\Rightarrow P_1(A) = P_2(A) \quad \forall A \in \sigma(\mathcal{F})$.

Ques 5. Let $\Omega = \{u_1, u_2, u_3, \ldots\}$ be the enumeration set such that

$\mathcal{F} = \{$field of all finite subsets and their complements$\}$

&

$\mu : \mathcal{F} \to \{0, 1\}$

$$\mu(A) = \begin{cases} 0, & \text{if } A \text{ is finite} \\ 1, & \text{if } A^c \text{ is finite.} \end{cases}$$

Now,

Let $A_1, A_2, \ldots A_n \in \mathcal{F}$

Case I: If all $\{A_i\}$ are finite & disjoint

$\Rightarrow \mu(A_i) = 0 \quad \forall i = 1, 2, \ldots n$

also $\left(\bigcup_{i=1}^{n} A_i\right)$ is finite

So $\mu\left(\bigcup_{i=1}^{n} A_i\right) = 0$

$\Rightarrow \mu\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} \mu(A_i)$.

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Case II

Now, but atleast, say A_j is infinite so that A_j^c is finite.

Now, to be $A_1, A_2, \ldots, A_j, \ldots A_n$ disjoint

$\{A_i\}_{i=\{1,2,\ldots j-1, j+1, \ldots n\}} \in A_j^c$.

$\Rightarrow \{A_i\}$ is finite $\forall\ i \in \{1, 2, \ldots j-1, j+1, \ldots n\}$

Now:

$\overset{n}{\underset{i=1}{\cup}} A_i$ is infinite

$\Rightarrow \mu\left(\overset{n}{\underset{i=1}{\cup}} A_i\right) = 1$

also $\mu(A_i) = \begin{cases} 0 & \text{if } i \in \{1, 2, \ldots j-1, j+1, \ldots n\} \\ 1 & \text{if } i = j \end{cases}$

So

$$\mu\left(\overset{n}{\underset{i=1}{\cup}} A_i\right) = \overset{n}{\underset{i=1}{\sum}} \mu(A_i)$$

$\Rightarrow \mu$ is finitely additive.

Now,

but $A_i = \{H_i\}$

$\Rightarrow \mu(A_i) = 0 \quad \forall\ i \in \mathbb{N}$

& $\left(\overset{\infty}{\underset{i=1}{\cup}} A_i\right) = \Omega$; where $\Omega^c = \phi$ finite

So

$$\mu(\Omega) = \mu\left(\overset{\infty}{\underset{i=1}{\cup}} A_i\right) = 1$$

&

$$\mu(\cdot) \neq \overset{\infty}{\underset{i=1}{\sum}} \mu(A_i) = 0$$

Hence μ is NOT Countbly additive on F.

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